1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Expanded hypothesis: In breast cancer, TP53 loss-of-function together with reduced E-cadherin (CDH1) expression drives EMT programs that promote stromal remodeling, resulting in an elevated stromal-to-tumor ratio (TSR) on H&E histology. This EMT-linked TSR elevation will be enriched in TP53-mutant, CDH1-low tumors and will independently predict worse overall survival (OS) after adjusting for clinicopathologic covariates. Mechanistically, TP53 dysfunction and CDH1 loss activate EMT transcriptional programs, weaken epithelial barriers, and recruit/expand stromal components, yielding higher TSR as a histologic readout of invasive potential. The association is expected to correlate with EMT gene signatures and to be amplified by inflammatory microenvironmental signals. Experimentally, in vitro and in vivo models with TP53 disruption and CDH1 suppression should exhibit enhanced invasion and extracellular matrix remodeling, supporting a causal link. The hypothesis is testable through (i) retrospective, multi-cohort analyses quantifying TSR and integrating TP53/CDH1 status with OS and EMT signatures, and (ii) functional assays validating EMT-driven invasion in controlled systems.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, CDH1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 dysfunction and loss of E-cadherin (CDH1) as the core molecular axis because both are mechanistically linked to EMT, epithelial barrier breakdown, and increased invasiveness; these connect directly to a stromal remodeling phenotype observable as an elevated stromal-to-tumor ratio on H&E.

CDH1: CDH1) as the core molecular axis because both are mechanistically linked to EMT, epithelial barrier breakdown, and increased invasiveness; these connect directly to a stromal remodeling phenotype observable as an elevated stromal-to-tumor ratio on H&E.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EMT	Supporting modifier from knowledge graph

Secondary	Klebsiella	Supporting modifier from knowledge graph
Secondary	CIS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of large breast cancer cohorts (e.g., TCGA-BRCA, METABRIC) with available WSIs, genomic data (TP53 mutation status), and transcriptomics to compute a Hallmark EMT score and CDH1 expression;

Prediction 2: digital pathology to quantify TSR (standardized pipeline with intra- and inter-observer validation);

Prediction 3: multivariate Cox models to test independent association of TSR with OS in TP53-mutant, CDH1-low tumors, adjusting for stage, grade, and receptor subtype;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- EMT: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Klebsiella: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.